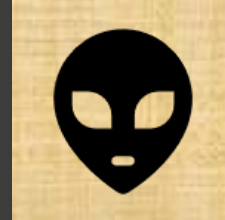




Uncovering Patterns in UFO Sightings

Topic Modeling and Network Analysis of Encounter Descriptions



INTRODUCTION

UFO sighting reports contain many real stories written by people who saw strange things in the sky.

We wanted to understand two main things:
What common themes appear in these reports?
How are the words and ideas connected to each other?

We used two methods:

Topic Modeling

This finds groups of words that often appear together. It helps discover hidden themes such as "bright lights in the sky" or "triangular objects".

Network Analysis

This shows how words are connected to each other. It creates a network where strongly related words are linked together.

Why these two?

Topic Modeling shows the main topics.

Network Analysis shows how the words inside those topics are connected.

Together they give a clear picture of the patterns in UFO reports.

DATA

We used thousands of real UFO sighting reports.

Each report has a long description column written by eyewitnesses.

This gives more than 1,000 text documents with categories (ufo_shape). Text Preprocessing

We cleaned the descriptions by:

- Making all letters small
- Removing punctuation and numbers
- Removing common words like "the", "and", "of"
- Changing words to their simple form

LIS 4/5693: Information Retrieval and Text Mining

Nathan Bratzler / Diego Macedo

University of Oklahoma

METHODOLOGY

Topic Modeling

Used Bag-of-Words and Latent Dirichlet Allocation (LDA) to find recurring patterns of co-occurring words.

Network Analysis

Built a word co-occurrence network to show connections between words that appear together in the same report.

Built a model to predict UFO shape from the description text using TF-IDF (Term Frequency-Inverse Document Frequency),

FINDINGS

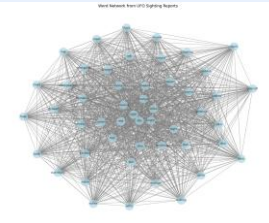


Figure 1: Word Network

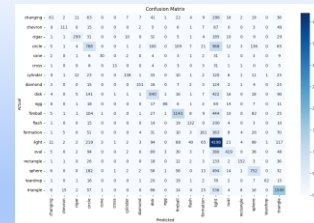


Figure 2: Confusion Matrix

CONCLUSION

- Topic modeling revealed clear themes such as bright lights and triangular shapes.
- Network analysis showed strong connections between words like "light", "sky", and "object".
- The machine learning model could predict some UFO shapes from the text. Text mining helps us find meaningful patterns in large collections of UFO stories.

REFERENCES

- Chapter 5: Text Clustering and Topic Modeling
- Text Pre-Processing chapter
- Text Visualization module (Altair)

GitHub Repository

Dataset

<https://shorturl.at/46Klb>

MODEL PERFORMANCE

Metric	Score
Model Used	Logistic Regression
Accuracy	55.2%
Macro Average F1-Score	0.49
Weighted Average F1-Score	0.54
Total Test Samples	20,146
Number of UFO Shapes Predicted	19

The model correctly predicted the UFO shape 55% of the time using only the text description.